

Zvolme $B = (2, x, x^2 + 2x + 2)$. B je patrně
báze prostoru V . Napišme si $[\phi]_B^B$: z definice

$$[\phi]_B^B = \left(\begin{array}{c|c|c} [\phi(2)]_B & [\phi(x)]_B & [\phi(x^2 + 2x + 2)]_B \end{array} \right)$$
$$= \left(\begin{array}{c|c|c} [0]_B & [2]_B & [2x^2 + 4x + 4]_B \end{array} \right) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Nyní

$$[\phi]_B^B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} J_{0,2} & 0 \\ 0 & J_{2,1} \end{pmatrix} = J$$

je matice v Jordanově tvaru.

□